

Self-governed multi-agent system for identifying early investment opportunities: HW3 Evolution

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I. INTRODUCTION

In this homework, we evolved **VibeInvestor** from a functional RAG agent into a robust agentic system. The system is designed to identify early-stage investment opportunities through a disciplined *Observe* $\rightarrow$ *Reason* $\rightarrow$ *Decide* $\rightarrow$ *Act* loop. For HW3, we focused on three key pillars: **Persistent State** (FAISS), **Role-based Orchestration** (Governor-Finder-Verifier), and **Adaptive Control** (Metric-driven feedback loops).

II. TOOLING RATIONALE

VibeInvestor integrates specialized tools to fulfill its research goals:

- **Google Gemini 2.0 Flash**: Acts as the reasoning engine for all roles.
- **SearchAPLio & yfinance**: Provide real-time web discovery and raw financial data.
- **FAISS Vector Database**: Stores historical research for semantic retrieval, enabling multi-session continuity.
- **RecursiveCharacterTextSplitter**: Ensures data ingestion stays within context windows while preserving semantic meaning.

III. ARCHITECTURE EVOLUTION (HW2 TO HW3)

The transition from HW2 to HW3 represents a shift from *linear execution* to *closed-loop orchestration*.

A. HW2 Foundation: Linear RAG

In HW2, the system operated as a single-agent pipeline. It would search, retrieve, and summarize in one pass. This led to "brittleness"; if the search results were poor, the final report was poor, with no opportunity for the system to self-correct.

B. HW3 Advancement: Multi-Agent Governance

For HW3, we decoupled responsibilities into distinct roles:

- 1) **The Finder (Explorer)**: Focused on maximizing recall from external tools.
- 2) **The Governor (Decision Maker)**: Acts as the "Executive," synthesizing the Finder's data and historical context from FAISS to make a final selection.
- 3) **The Verifier (Critic)**: A new, independent role that calculates a **Groundedness Score**. It ensures the Governor did not "hallucinate" metrics or names that weren't in the Finder's raw output.

This modularity allows the system to fail gracefully; if the Verifier detects low groundedness, the ****Adaptive Loop**** triggers a retry with modified strategies.

IV. ADAPTIVE CONTROL & MEMORY LOGIC

We implemented persistent state using ****FAISS****. Short-term state (feedback) is passed between iterations in a `while` loop, while long-term state (past winners) is retrieved during the Governor's reasoning phase. The system adapts by relaxing constraints (e.g., P/E ratios) if the initial search yields insufficient data.

V. METRICS & RESULTS

The following table summarizes the performance of VibeInvestor across 5 automated test cases.

TABLE I
HW3 AUTOMATED EVALUATION RESULTS

Test Case	Attempts	Completion	Adherence	Groundedness
1. Baseline (Semiconductors)	2	1.0	0.70	1.00
2. Specificity (P/E $\leq$ 20)	4	1.0	0.00	0.60
3. Memory (Competitor Search)	2	1.0	0.95	1.00
4. Adaptive (Deep-Sea)	3	1.0	0.70	0.71
5. Failure (Teleportation)	N/A	0.0	0.00	0.00

VI. FAILURE CASE DEEP DIVE (CASE 2)

What happened: The system sought Healthcare AI companies with P/E $\leq$ 20. It ran for 4 iterations.

Why it happened: Strict valuation filters often fail on web-scraped data. The Governor prioritized "Groundedness" over "Adherence," selecting a well-documented stock (Micron) rather than a hallucinated healthcare stock.

How we fixed it: We implemented `robust_extract_findings` in `main.py` to handle Gemini's conversational JSON outputs.

VII. SCALABILITY REFLECTION

A. 1,000 Users

Local FAISS (`faiss_investment_db`) would suffer from write-locks. We would migrate to a cloud vector database like **Pinecone** and use a distributed task queue (e.g., Celery) to manage parallel agent runs.

B. High-Stakes & Regulated Industry

In finance, hallucinations are a legal risk. We would add a **Deterministic Guardrail** layer to verify tickers against official exchanges before a PDF is generated, and restrict the Finder to whitelisted domains (SEC filings, Bloomberg).

VIII. SUMMARY

VibeInvestor now demonstrates a complete agentic lifecycle. Through multi-agent orchestration and persistent memory, the system can self-correct and build a cumulative knowledge base for investment research.